CLI use

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References

1. An introduction to the CLI

CLI (Command-Line Interface, the command line interface) is a user interface with which you interact with a computer system or a program by typing text commands on the keyboard. It was the mainstream interaction method before the graphical user interface (GUI) became widespread, and it is still widely used today in the fields of system administration, automation scripts, development tools and so on.

2. Enable the maximum performance of the mainboard

2.1. Enable the MAX power mode

Enabling the MAX Power Mode (the maximum power mode) on the Jetson ensures that all the CPU and GPU cores are turned on:

```
sudo nvpmodel -m 0
```

2.2. Enable the Jetson clocks

Enabling the Jetson Clocks ensures that all the CPU and GPU cores run at the maximum frequency:

```
sudo jetson_clocks
```

3. Predict with the CLI

The following CLI commands are all run in the desktop terminal.

3.1. CLI usage syntax

```
yolo TASK MODE ARGS

Where   TASK (optional) is one of [detect, segment, classify, pose, obb]
        MODE (required) is one of [train, val, predict, export, track, benchmark]
        ARGS (optional) are any number of custom 'arg=value' pairs like 'imgsz=320' that
override defaults.
```

3.2. Image prediction

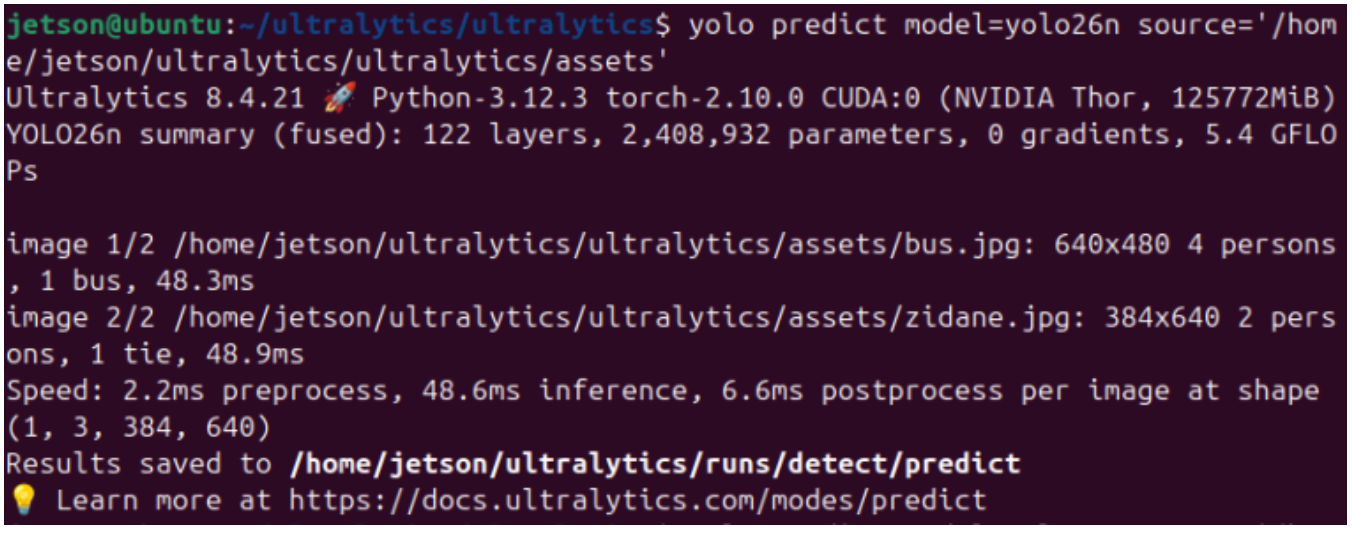
Use yolo26n.pt to predict the images that come with the ultralytics project: if the system does not find the corresponding model file in the directory in which the command is run, it is downloaded automatically (if it cannot be downloaded, you can copy the model in yourself)

Enter the project folder:

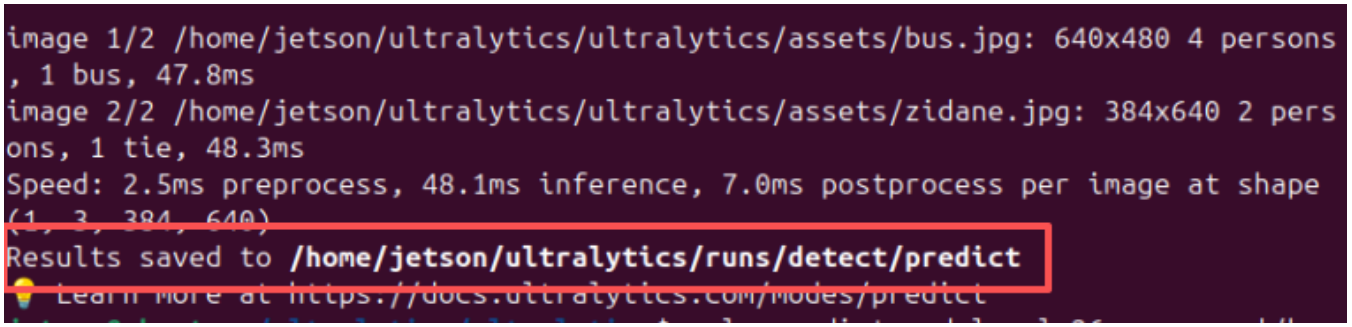
```
cd /home/jetson/ultralytics/ultralytics/
```

Use yolo26n.pt to detect the images in the target folder and output the results:

```
yolo predict model=yolo26n source='/home/jetson/ultralytics/ultralytics/assets'
```



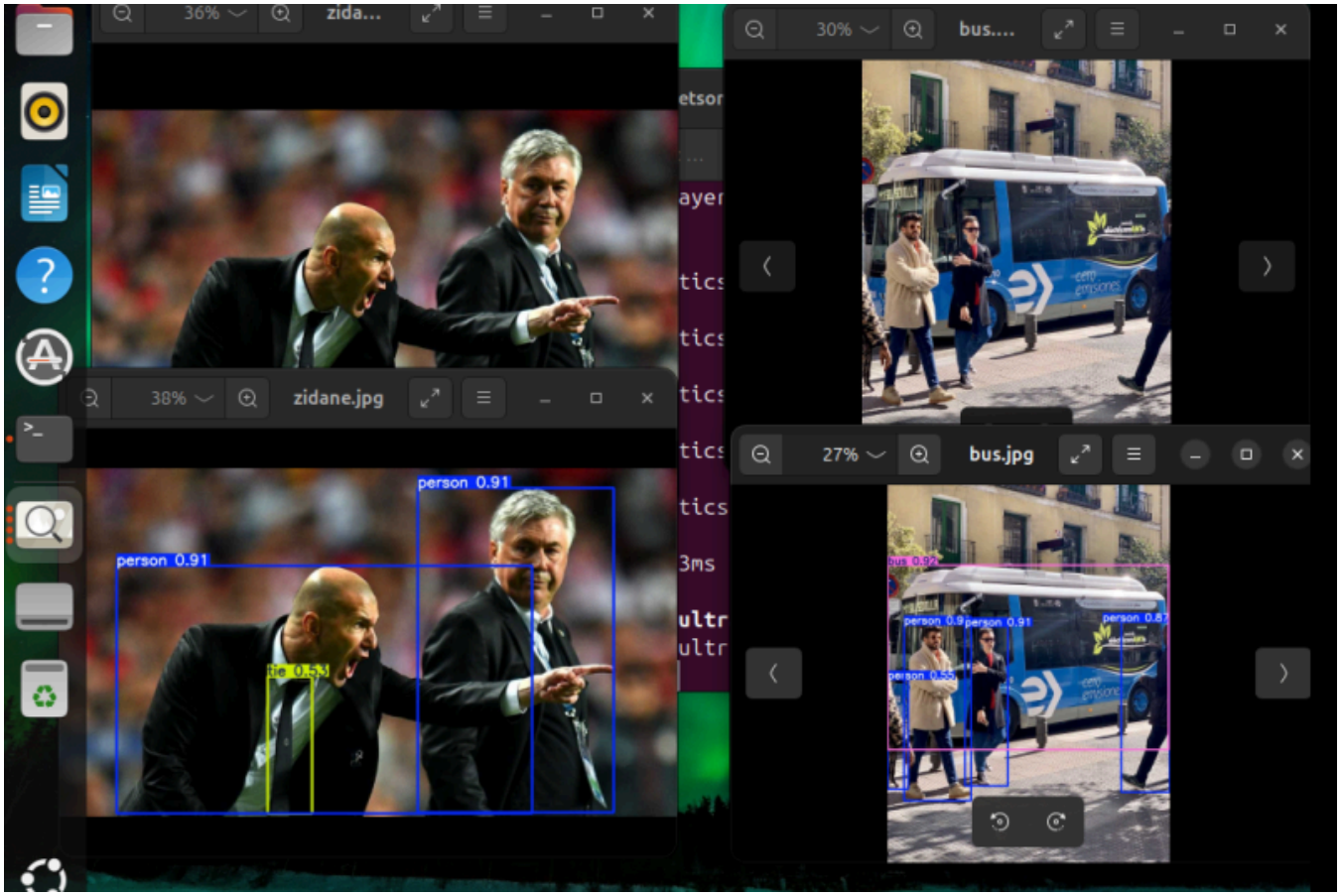
Note: every time it runs, the predict under the output folder is incremented automatically by 1; note down the path that the program prints.



Effect preview

The location of the output of the yolo recognition: /home/jetson/ultralytics/runs/detect

Note: the actual path is the one printed above. /home/jetson/ultralytics/runs/detect/predict



3.3. Video prediction

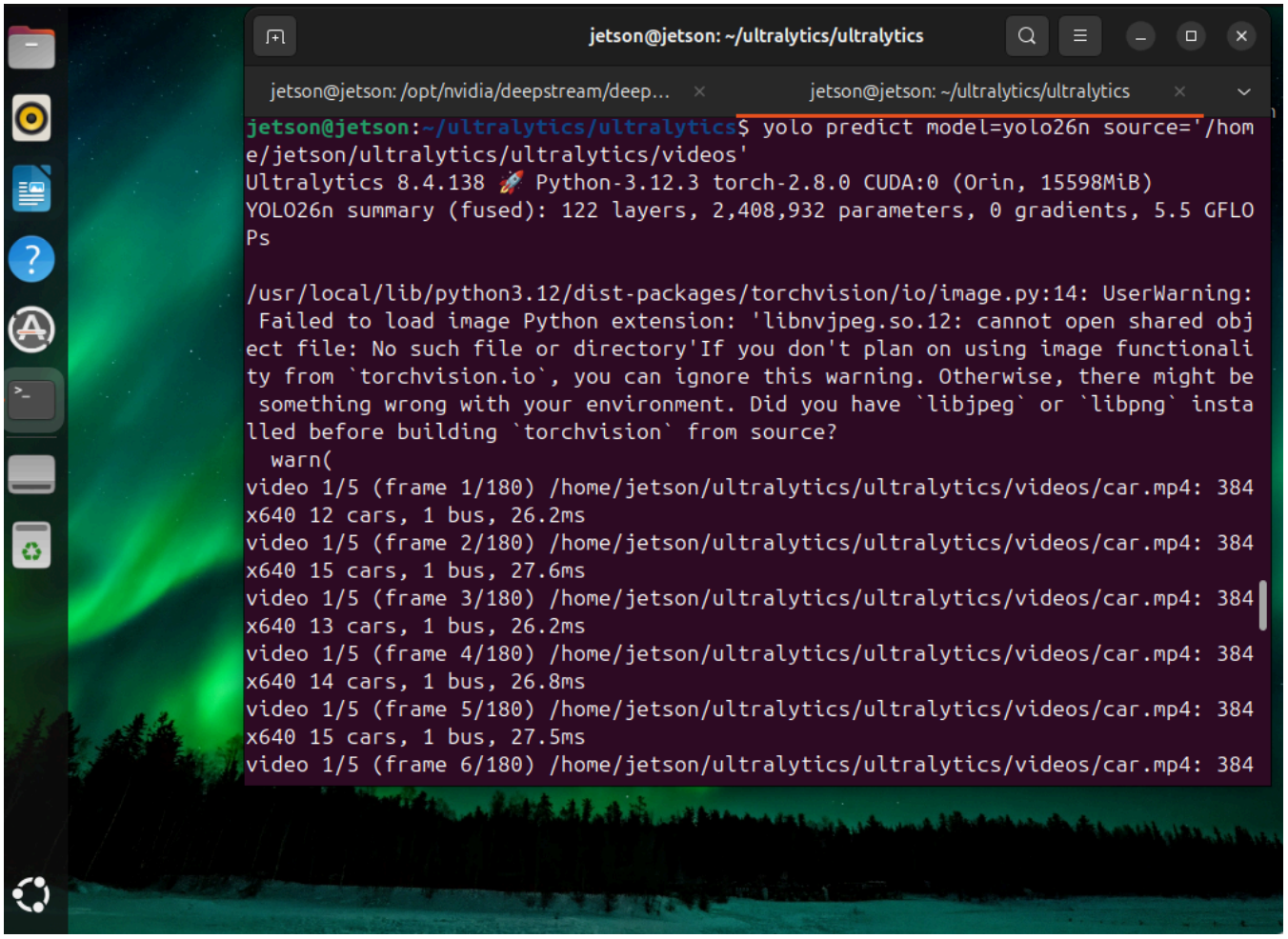
Use yolo26n.pt to predict the videos under the ultralytics project (not the videos that come with ultralytics; if the program cannot find the video, you have to put the video file manually into the /home/jetson/ultralytics/ultralytics/videos path): if the system does not find the corresponding model file in the directory in which the command is run, it is downloaded automatically (if it cannot be downloaded, you can copy the model in yourself)

Enter the project folder:

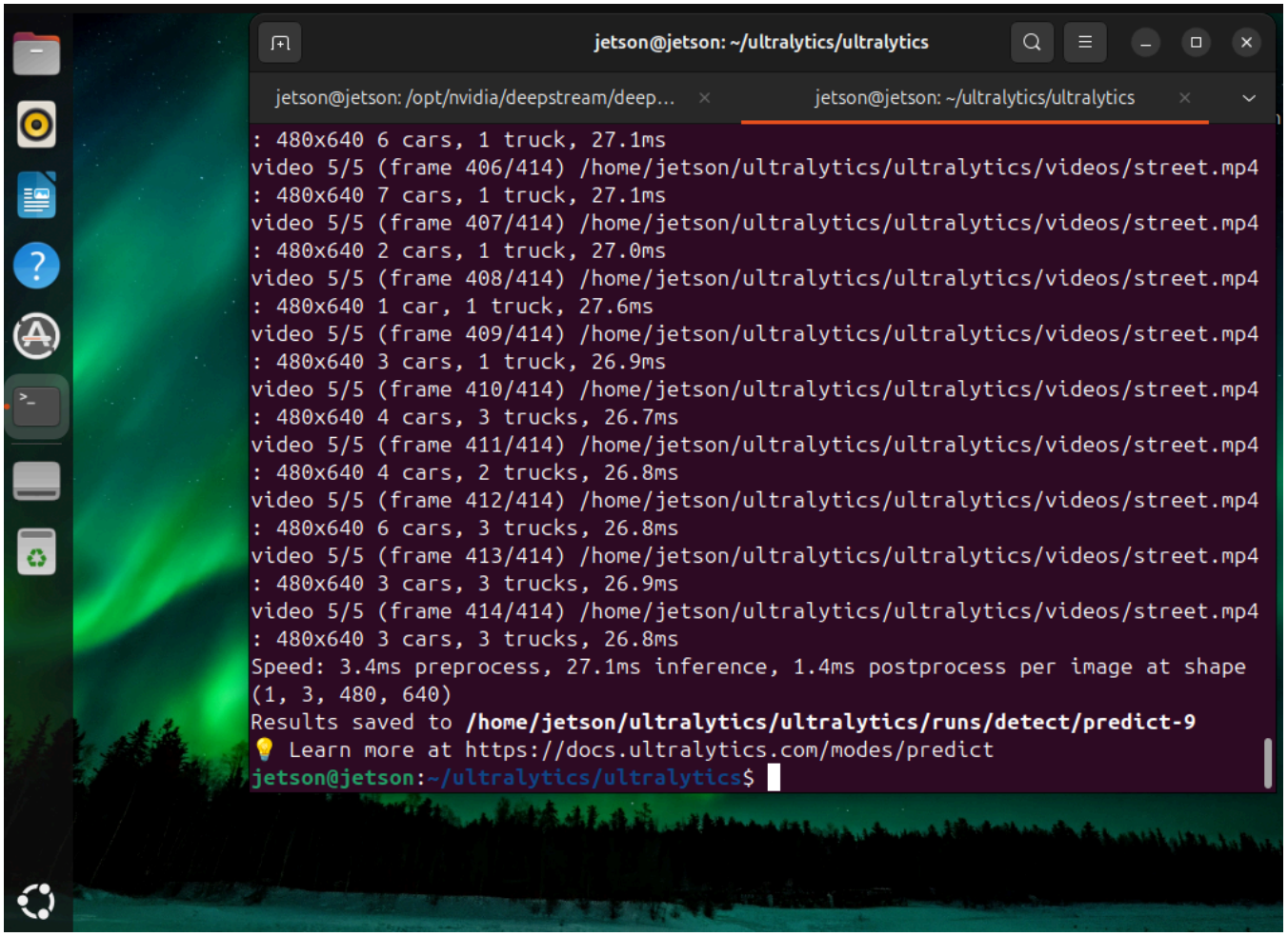
```
cd /home/jetson/ultralytics/ultralytics
```

Use yolo26n.pt to detect the videos in the target folder and output the results:

```
yolo predict model=yolo26n source='/home/jetson/ultralytics/ultralytics/videos'
```



Every time it runs, the predict under that folder is incremented automatically by 1; note down the path that the program prints.



Preview of the effect

The location of the videos that the yolo recognition outputs: /home/jetson/ultralytics/runs/detect

You can watch the video with ffmpeg; install it with the following command:

```
sudo apt install ffmpeg -y
```

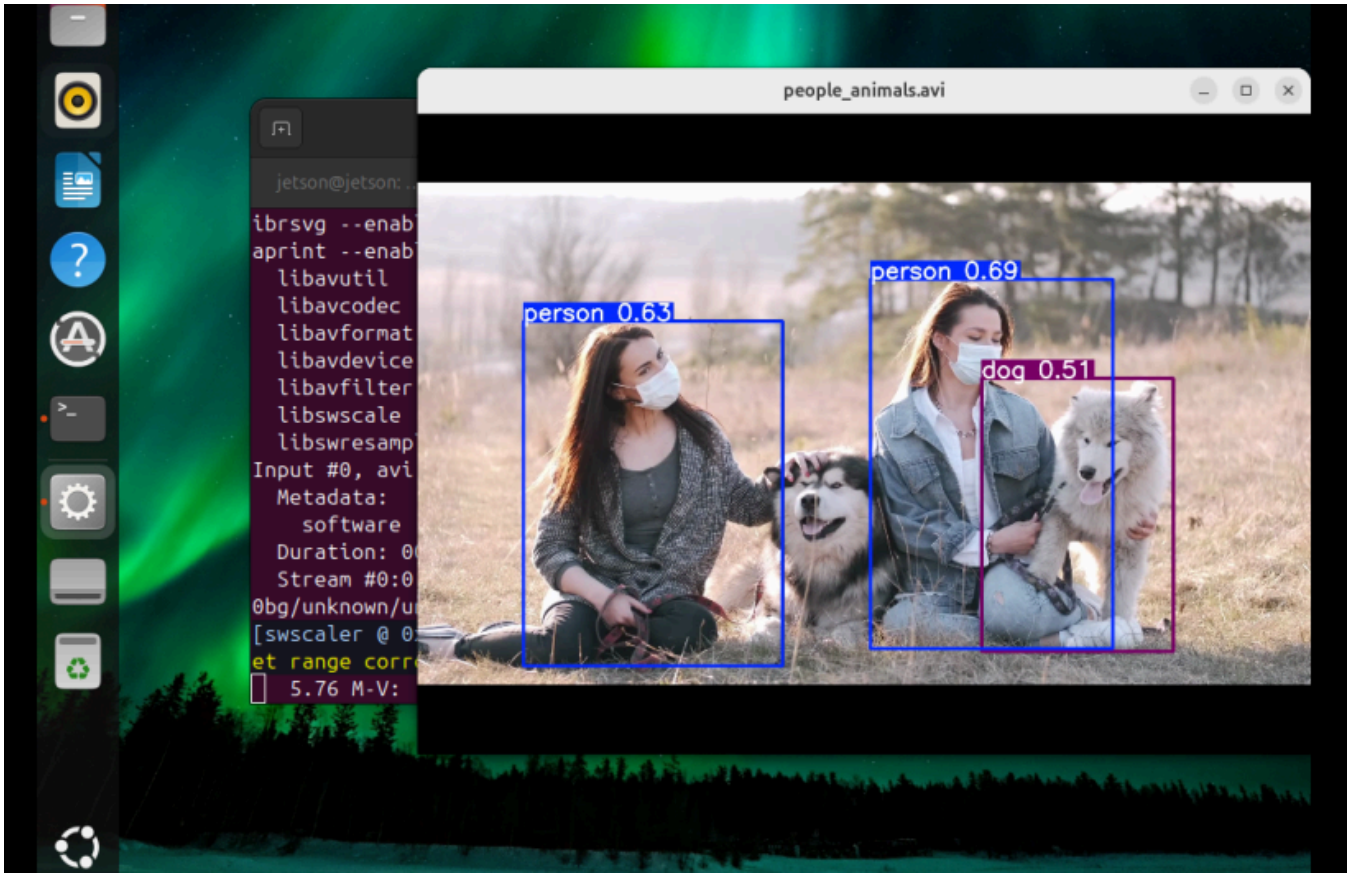
Enter the path of the video input:

Note: because the predict is incremented automatically by 1 every time the program runs, the actual output is the reference. predict-9 has to be changed to the actual name of the folder.

```
cd /home/jetson/ultralytics/ultralytics/runs/detect/predict-9
```

Play the video:

```
ffplay people_animals.avi
```



3.4. Real-time prediction

First plug the USB camera into the USB port of the mainboard.

Use yolo26n.pt to predict the image of the USB camera: if the system does not find the corresponding model file in the directory in which the command is run, it is downloaded automatically (if it cannot be downloaded, you can copy the model in yourself)

Enter the project folder:

```
cd /home/jetson/ultralytics/ultralytics
```

Use yolo26n.pt to detect the camera image and output the results; choose one of the following functions to run.

After the program runs, it opens the camera image, detects it in real time and displays the output result on the image.

```
# Object detection
yolo predict model=yolo26n.pt source=0 save=False show

# Instance segmentation:
yolo predict model=yolo26n-seg.pt source=0 save=False show

# Image classification:
yolo predict model=yolo26n-cls.pt source=0 save=False show
```

```
# Pose estimation:
yolo predict model=yolo26n-pose.pt source=0 save=False show

# Oriented object detection:
yolo predict model=yolo26n-obb.pt source=0 save=False show
```

If it is the first time you run the program, the model has to be loaded, so the loading takes a while; please be patient.

Click in the terminal and press the "Ctrl + C" shortcut to terminate the program!

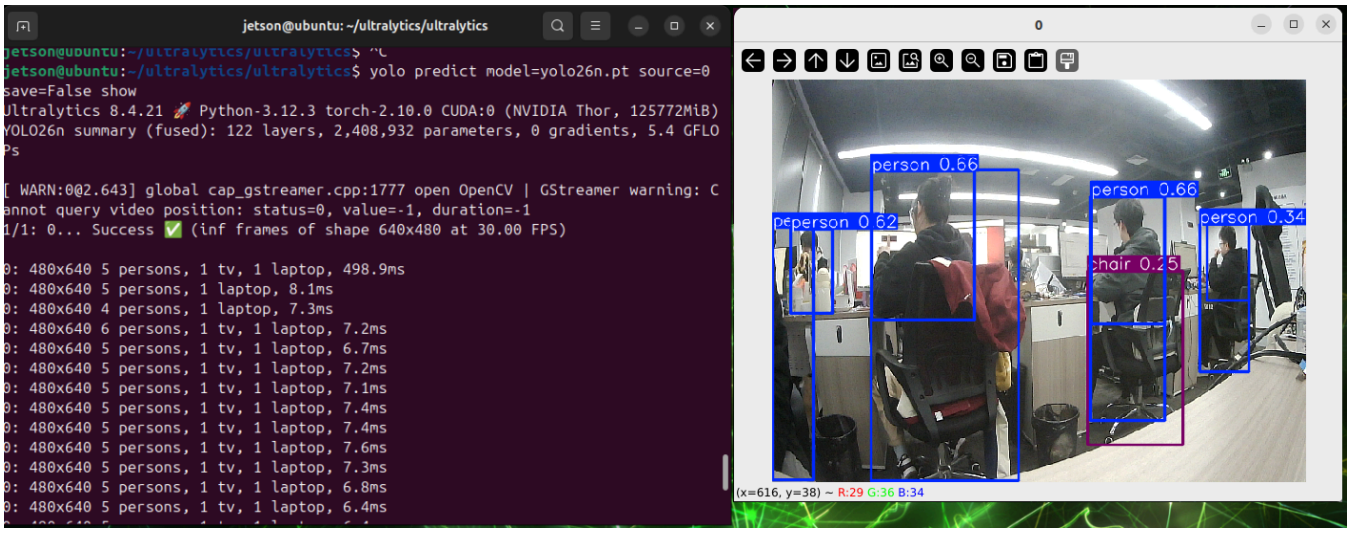
> **Parameter description**

`model`: specifies the YOLO model

`source`: specifies the recognition source: the number can be switched among multiple cameras

`save=False`: disables saving the results

`show`: displays in real time



References

https://github.com/ultralytics/ultralytics

https://docs.ultralytics.com/usage/cli/